THE UNIVERSITY OF MELBOURNE

Student Number

Semester 1 Assessment, 2020

School of Mathematics and Statistics

MAST10012 Introduction to Mathematics

This exam consists of 21 pages (including this page)

Authorised materials: printed one-sided copy of the Exam or the Masked Exam made available earlier (or an offline electronic PDF reader), handwritten notes, and blank A4 paper

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Instructions to Students

- During exam writing time you may only interact with the device running the Zoom session with supervisor permission. The screen of any other device must be visible in Zoom from the start of the session.
- If you have a printer, print out the exam single-sided and hand write your solutions into the answer spaces.
- If you do not have a printer, or if your printer fails on the day of the exam,

  (a) download the exam paper to a second device (not running Zoom), disconnect it from the internet as soon as the paper is downloaded and read the paper on the second device;

  (b) write your answers on the Masked Exam PDF if you were able to print it single-sided before the exam day.

  If you do not have the Masked Exam PDF, write single-sided on blank sheets of paper.
- If you are unable to answer the whole question in the answer space provided then you can append additional handwritten solutions to the end of your exam submission. If you do this you MUST make a note in the correct answer space or page for the question, warning the marker that you have appended additional remarks at the end.
- Assemble all the exam pages (or template pages) in correct page number order and the correct way up, and add any extra pages with additional working at the end.
- Scan your exam submission to a single PDF file with a mobile phone or a scanner. Scan from directly above to avoid any excessive keystone effect. Check that all pages are clearly readable and cropped to the A4 borders of the original page. Poorly scanned submissions may be impossible to mark.
- Upload the PDF file via the Canvas Assignments menu and submit the PDF to the GradeScope tool by first selecting your PDF file and then clicking on Upload PDF.
- Confirm with your Zoom supervisor that you have GradeScope confirmation of submission before leaving Zoom supervision.
- You should attempt all questions.
- There are 10 questions with marks as shown. The total number of marks available is 100.

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Question 1 (27 marks)

Short-answer questions: Full marks will be given for a correct answer placed in the box. Part marks will be given for incorrect answers if working is shown.

(a) Evaluate $\cos\left(-\frac{\pi}{3}\right) + \sin\left(\frac{7\pi}{2}\right)$

(b) An Italian restaurant is offering a special degustation deal for large groups. They offer any combination of 5 different pizzas and 5 different pastas for \$150. Their menu has 15 different pizzas and 10 different pasta options to choose from. How many different degustation menus can the customers make up?

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(c) Let $f(x) = (x^2 + 4x + 2)\sin(x)$. Find $f'(x)$.

(d) Let $g(x) = \dfrac{e^x}{\sqrt{x}}$. Find $g'(x)$.

(e) Let $h(x) = (\log(6x))^3$. Find $h'(x)$.

(f) Find the equation of the tangent line of $f(x) = (\cos(x) + 1)^2$ about $x = \frac{\pi}{2}$

(g) Evaluate the following:

$$\int \frac{3x^5 + 2x}{x^2}\, dx$$

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(h) Evaluate the following:

$$\int_0^8 \frac{1}{\sqrt[3]{x^2}} + \cos(\pi x)\, dx$$

(i) Solve the following initial value problem

$$F(x) = \int e^{-5x}\, dx \quad \text{if } F(0) = 0$$

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Full-solution problems: Complete solutions are required for the remainder of this examination. Be sure to justify your answers and to show all your work. In addition to the accuracy of your answer and methods, you will be assessed on the justification of your solutions.

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Question 2 (7 marks)

Find all solutions of the trigonometric equation $2\cos(6x) + \sqrt{3} = 0$.

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Question 3 (7 marks)

(a) Evaluate the following limit if it exists. Otherwise, write DNE and justify why the limit does not exist.

$$\lim_{x \to 2} \frac{x^3 - 9x^2 + 26x - 24}{x - 2}$$

(b) Use the limit definition of the derivative to find $f'(x)$ given $f(x) = \frac{1}{2x}$ for $x \neq 0$. No marks will be awarded for using any other method.

Question 4 (10 marks)

Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = -x^2 + 8x - 15$.
Hint: You may use, without proof, that the function can also be written as

$$f(x) = -x^2 + 8x - 15 = (3 - x)(x - 5) = -(x - 4)^2 + 1$$

(a) What is the domain, codomain, and range of f?

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(b) Find the largest value q such that the function $f : (-\infty, q] \to \mathbb{R}$ given by

$$f(x) = -x^2 + 8x - 15$$

is injective.

(c) Find the interval C such that $f : (-\infty, q] \to C$ given by

$$f(x) = -x^2 + 8x - 15$$

is surjective.

(d) Based on your choice of q and C, explain why the function $f : (-\infty, q] \to C$ given by $f(x) = -x^2 + 8x - 15$ is invertible.

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(e) Define g to be the inverse of f. State the domain and codomain of g as well as provide the algebraic equation for $g(x)$.

Question 5 (11 marks)

Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = e^x(x^2 - 1)$.
Hint: It may be useful to know that $\sqrt{2} \approx 1.41$

(a) Find all x and y intercepts of the function.

(b) Find the intervals where $f(x)$ is positive and intervals where $f(x)$ is negative

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(c) Evaluate and simplify $f'(x)$

(d) Find the points in the domain where $f'(x) = 0$.

(e) Find the intervals where $f(x)$ is increasing and intervals where $f(x)$ is decreasing

(f) Find and classify the location of any local extrema of the function.

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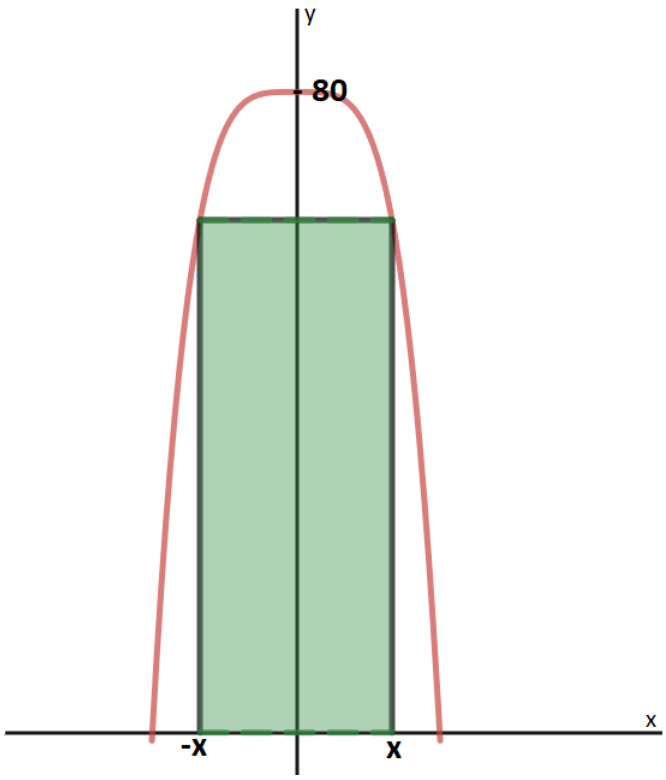
Question 6 (11 marks)

The following graph shows the function $y = -x^4 + 80$ with a general rectangle drawn with its base along the x axis



Hint: You might find useful to know that $\sqrt[4]{80} \approx 2.99$ and $\sqrt[4]{16} = 2$.

(a) What is the length of the base of the rectangle in terms of x?

(b) Given that the upper vertices of the rectangle touch the curve, what is the height of the rectangle in terms of x?

(c) Write down an expression for the Area of the rectangle in terms of x. What is the domain of $A(x)$.

(d) Determine the dimensions (length, height and area) of the rectangle with the maximum area that will fit under this curve.

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Question 7 (10 marks)

Consider two functions given by

$$y = x^2 + 1 \qquad \text{and} \qquad y = x + 3$$

(a) Find the coordinates where the two functions intersect.

(b) Find the **physical** area between the functions f and g.

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Question 8 (5 marks)

Consider the discrete probability distribution for the random variable X shown in the table below:

x	-1	0	1	2
$P(X = x)$	0.1	a	0.4	0.3

(a) Find the value of a

(b) Find $\mathbb{E}(X)$

(c) Find the probability that $X = 2$, given that $X \geq 1$

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Question 9 (7 marks)

The weight of Australian avocados is normally distributed with mean 200g and standard deviation 10g.

(a) Suppose that we sell Australian avocados at three different prices, depending on the size (small, medium or large). If the bottom 10% of avocados by weight are sold at the price of "small" avocados. What is the maximum weight of a "small" avocado? Your answer should be accurate to 1 decimal place.

(b) Avocados sold at the price of "large" avocados must weight at least 215g. What percentage of avocados can be sold at that price?

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Question 10 (5 marks)

Careful reasoning is required for this question. Even the right answer with insufficient justification will not score well in this question. When rolling two dice, find the probability of getting a greater number on the first die than the one on the second one, given that the sum should equal 8.

End of exam questions—Total Available Marks = 100

Turn the page for appended material

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Formula Sheet

- **Combinatorics:**

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot 3 \cdot 2 \cdot 1$$
$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

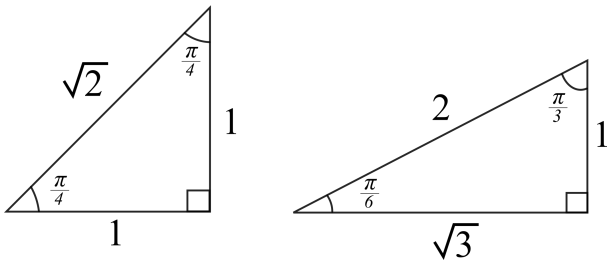
- **Functions:**
  - Lines

$$y = mx + c \qquad\qquad y - b = m(x-a)$$

  - Quadratics

$$y = ax^2 + bx + c = a(x-r)(x-s) = a(x-h)^2 + k$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **Special Triangles:**



- **Trigonometric identities:**

$$\sin(x + 2\pi) = \sin(x) \qquad \cos(x + 2\pi) = \cos(x)$$
$$\sin(\pi - x) = \sin(x) \qquad \sin(-x) = -\sin(x)$$
$$\cos(\pi - x) = -\cos(x) \qquad \cos(-x) = \cos(x)$$

$$\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y) \qquad \cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$$
$$\sin(2x) = 2\sin(x)\cos(x) \qquad \cos(2x) = \cos^2(x) - \sin^2(x)$$
$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \qquad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

- **Exponential and logarithms:**

$$a^0 = 1 \qquad a^1 = a$$
$$a^{-1} = \frac{1}{a} \qquad a^{\frac{1}{2}} = \sqrt{a}$$
$$a^x \cdot a^y = a^{x+y} \qquad \frac{a^x}{a^y} = a^{x-y}$$
$$(a^x)^y = a^{x \cdot y} \qquad \sqrt[y]{a^x} = a^{\frac{x}{y}}$$
$$a^x \cdot b^x = (ab)^x \qquad \frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$$

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$$\log_e(x) = \ln(x) \qquad \log_a(1) = 0$$
$$\log_a(a) = 1 \qquad \log_a(p^x) = x\log_a(p)$$
$$\log_a(pq) = \log_a(p) + \log_a(q) \qquad \log_a\left(\frac{p}{q}\right) = \log_a(p) - \log_a(q)$$
$$\log_a(p) = \frac{\log_c(p)}{\log_c(a)}$$

- **Limits:** If we have

$$\lim_{x\to a} f(x) = K \qquad \lim_{x\to a} g(x) = L$$

Then

$$\lim_{x\to a}[f(x)+g(x)] = K + L \qquad \lim_{x\to a}[f(x)-g(x)] = K - L$$
$$\lim_{x\to a}[f(x)\times g(x)] = K \times L \qquad \lim_{x\to a}\left[\frac{f(x)}{g(x)}\right] = \frac{K}{L}$$
$$\lim_{x\to a}[f(x)]^n = \left[\lim_{x\to a} f(x)\right]^n \qquad \lim_{x\to a}[f(x)]^{\frac{1}{n}} = \left[\lim_{x\to a} f(x)\right]^{\frac{1}{n}}$$
$$\lim_{x\to a} cf(x) = cK$$
$$\lim_{x\to a} c = c \qquad \lim_{x\to a} x = a$$
$$\lim_{x\to a} x^n = a^n \qquad \lim_{x\to a} x^{\frac{1}{n}} = a^{\frac{1}{n}}$$

- **Derivatives:**

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{a\to x}\frac{f(a)-f(x)}{a-x}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(c\Big) = 0 \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(x^k\Big) = kx^{k-1}$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\sqrt{x}\Big) = \frac{1}{2\sqrt{x}} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(e^x\Big) = e^x$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\log(x)\Big) = \frac{1}{x} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(\sin(x)\Big) = \cos(x)$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\cos(x)\Big) = -\sin(x) \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(\tan(x)\Big) = \frac{1}{\cos^2(x)}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(u(x)\cdot v(x)\Big) = \frac{\mathrm{d}u}{\mathrm{d}x}\cdot v(x) + u(x)\cdot\frac{\mathrm{d}v}{\mathrm{d}x} = u'(x)\cdot v(x) + u(x)\cdot v'(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u(x)}{v(x)}\right) = \frac{\frac{\mathrm{d}u}{\mathrm{d}x}\cdot v(x) - u(x)\cdot\frac{\mathrm{d}v}{\mathrm{d}x}}{[v(x)^2]} = \frac{u'(x)\cdot v(x) - u(x)\cdot v'(x)}{[v(x)^2]}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(f(u(x))\Big) = \frac{\mathrm{d}f}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}x} = f'(u)\cdot u'(x)$$

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The standard normal distribution table for $P(X < x)$

x	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000